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(54) **MULTI-TUBE REACTOR SYSTEMS AND PROCESSES FOR NO-OXIDATIVE CONVERSION OF METHANE**

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(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,127,389 A * 11/1978 Hackemesser B01J 8/062 165/157

4,652,687 A 3/1987 Imai et al.
(Continued)

OTHER PUBLICATIONS

Zamara et al., Zeolite washcoating onto Cordierite honeycomb reactors for environmental applications, Chemical Engineering Journal, vol. 106, pp. 25-33, 2005.

(Continued)

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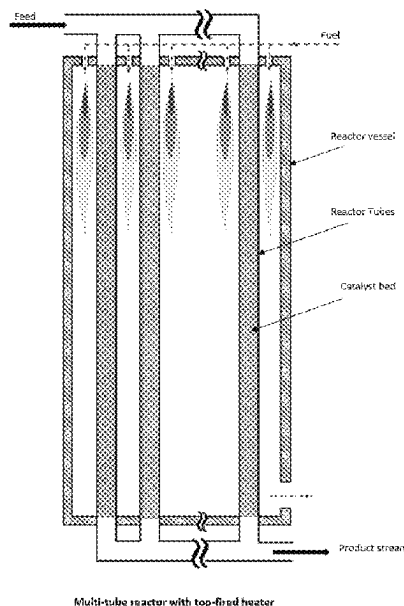
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(57) **ABSTRACT**

The present disclosure refers to systems and methods for efficiently converting a C₁-C₃ alkane such as natural gas to a liquid C₂-C₁₀ product and hydrogen. Generally, the process comprises flowing the C₁-C₃ alkane through a plurality of tubes within a vessel wherein the tubes house a catalyst for converting the C₁-C₃ alkane to the liquid C₂-C₁₀ product and hydrogen. The C₁-C₃ alkane is heated under suitable conditions to produce the liquid C₂-C₁₀ product and hydrogen. Advantageously, the C₁-C₃ alkane is heated by burning a fuel outside the tubes in fuel burning nozzles configured to transfer heat from the burning through the tubes.

13 Claims, 2 Drawing Sheets



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2011/0060176 A1 3/2011 Kiesslich et al.
2012/0165418 A1 6/2012 Park et al.
2016/0007484 A1 1/2016 Daley, III
2016/0289141 A1 10/2016 Bachmann et al.
2016/0362351 A1 12/2016 Nagaki et al.
2017/0007989 A1 1/2017 Coelho Tsou et al.
2021/0115341 A1 4/2021 Sampath
2021/0309589 A1 10/2021 Yun et al.

OTHER PUBLICATIONS

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,202,517 A 4/1993 Minet et al.
2002/0103402 A1 8/2002 Chang et al.
2009/0123348 A1* 5/2009 Brady F28F 13/14
423/650
2010/0312029 A1 12/2010 Gulotty, Jr.

Aboul-Gheit et al., Oxygen free conversion of natural gas to useful hydrocarbons and hydrogen over monometallic Mo and Bimetalli Mo—Fe, Mo—Co or Mo-Ni/HZSM-5 catalysts prepared by mechanical mixing, Fuel Processing Technology, vol. 102, pp. 24-29, May 14, 2012.
International Search REport and Written Opinion dated Jun. 16, 2023 issued in PCT/US2023/14280.

* cited by examiner

FIGURE 1

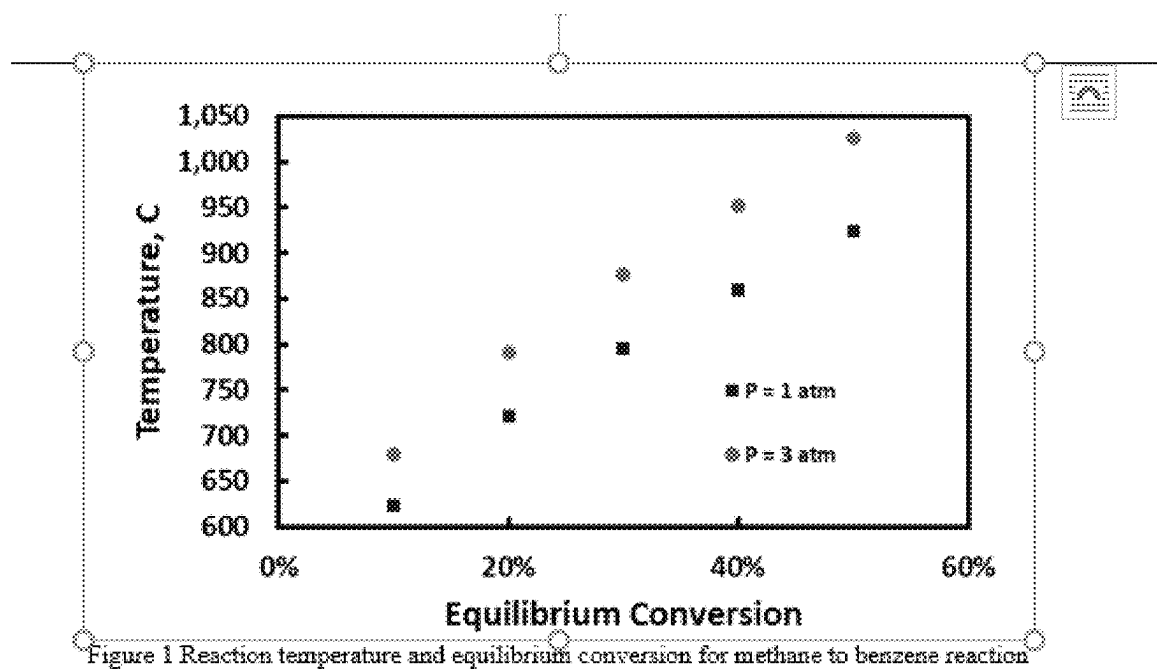
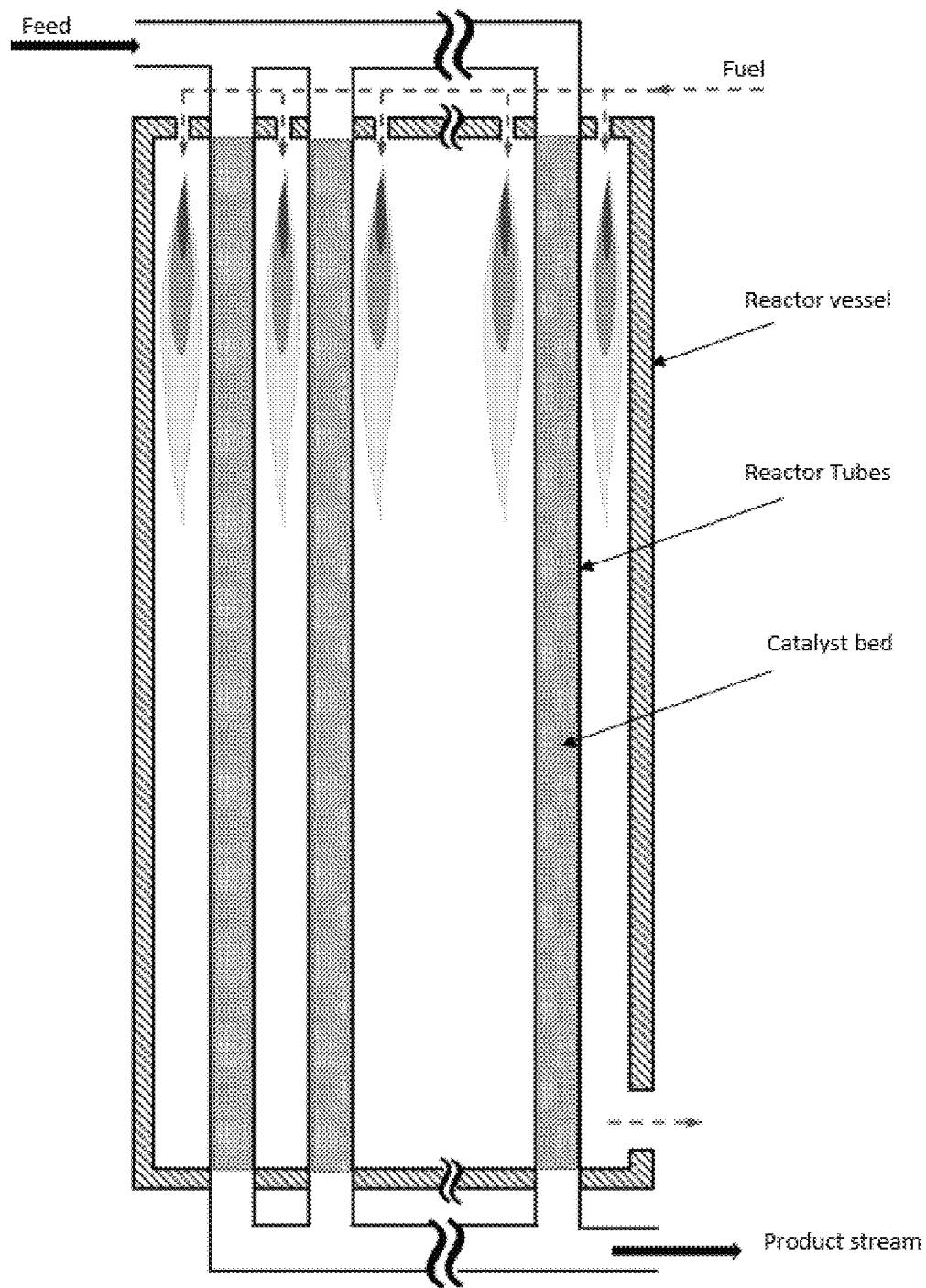


FIGURE 2

**Multi-tube reactor with top-fired heater**

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MULTI-TUBE REACTOR SYSTEMS AND PROCESSES FOR NO-OXIDATIVE CONVERSION OF METHANE

FIELD OF THE DISCLOSURE

The present disclosure relates to systems and methods for converting a C_1 - C_3 alkane to a liquid C_2 - C_{10} product and hydrogen.

BACKGROUND AND SUMMARY

Hydrogen is one of the more important options for future clean energy. In addition, higher molecular weight hydrocarbons such as olefins and aromatics are often desirable as value added chemicals. While methane has been used to make such value added chemicals in the past the process are usually not cost-effective. What is needed is a solution that produces hydrogen and value added chemicals in a cost-effective manner. It would further be advantageous if such a solution was not energy intensive.

Advantageously, the instant application pertains to new systems and methods that advantageously produce hydrogen and value added chemicals in a cost-effective manner and are not energy intensive.

In one embodiment the application pertains to a process for converting a C_1 - C_3 alkane to a liquid C_2 - C_{10} product and hydrogen. The process comprises flowing the C_1 - C_3 alkane through a plurality of tubes within a vessel wherein the tubes house a catalyst for converting the C_1 - C_3 alkane to the liquid C_2 - C_{10} product and hydrogen. The C_1 - C_3 alkane is heated under suitable conditions to produce the liquid C_2 - C_{10} product and hydrogen. The C_1 - C_3 alkane is heated by burning a fuel outside the tubes in fuel burning nozzles configured to transfer heat from the burning through the tubes.

In another embodiment the application pertains to a process for converting natural gas to a liquid C_2 - C_{10} product comprising ethylene, benzene, naphthalene, or a mixture thereof and hydrogen. The process comprises flowing the natural gas through a plurality of tubes within a vessel wherein the tubes house a catalyst for converting the natural gas to the liquid C_2 - C_{10} product comprising ethylene, benzene, naphthalene, or a mixture thereof and hydrogen. The natural gas is heated at a temperature of from about 500, or from about 700, up to about 1000 or up to about 1200° C., and a pressure of from about 1 atmosphere up to about 3, or up to about 5, or up to about 10, or up to about 20 atmospheres to produce the liquid C_2 - C_{10} product comprising ethylene, benzene, naphthalene, or a mixture thereof and hydrogen. The natural gas is heated by burning a fuel outside the tubes in fuel burning nozzles configured to transfer heat from the burning through the tubes. Additionally or alternatively, electric or other heating may be employed with the burning fuel or in place of it.

In another embodiment the application pertains to a reactor for conversion of alkanes to liquid hydrocarbons and hydrogen. The reactor comprises a vessel and a plurality of tubes within the vessel wherein the tubes are configured to house a catalyst. The vessel is configured to burn a fuel outside the plurality of tubes. The vessel is further configured to transfer heat from the burning fuel to the catalyst housed within the plurality of tubes.

These and other objects, features and advantages of the exemplary embodiments of the present disclosure will become apparent upon reading the following detailed

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description of the exemplary embodiments of the present disclosure, when taken in conjunction with the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments of the present disclosure, together with further objects and advantages, may best be understood by reference to the following description taken in conjunction with the accompanying drawings.

FIG. 1 shows reaction temperature, pressure, and equilibrium conversion for a methane to benzene reaction.

FIG. 2 shows a representative multi-tube reactor with top fired heater.

DETAILED DESCRIPTION

The following description of embodiments provides a non-limiting representative examples referencing numerals to particularly describe features and teachings of different aspects of the invention. The embodiments described should be recognized as capable of implementation separately, or in combination, with other embodiments from the description of the embodiments. A person of ordinary skill in the art reviewing the description of embodiments should be able to learn and understand the different described aspects of the invention. The description of embodiments should facilitate understanding of the invention to such an extent that other implementations, not specifically covered but within the knowledge of a person of skill in the art having read the description of embodiments, would be understood to be consistent with an application of the invention.

General Process

The instant application pertains to a process for converting a C_1 - C_3 alkane to a liquid C_2 - C_{10} product and hydrogen. The process generally comprises first flowing the C_1 - C_3 alkane through a plurality of tubes within a vessel. The tubes typically house a catalyst for converting the C_1 - C_3 alkane to the liquid C_2 - C_{10} product and hydrogen. The C_1 - C_3 alkane is not particularly limited and may include, for example, natural gas, methane, ethane, propane, or mixtures thereof. As used herein natural gas comprises methane and potentially higher alkanes, carbon dioxide, nitrogen or other gases, and/or sulfide compounds such as hydrogen sulfide, and mixtures thereof. The produced product typically comprises liquid C_2 - C_{10} product and hydrogen. The liquid C_2 - C_{10} product is not particularly limited and could be saturated, unsaturated, aromatic, or a mixture of such compounds. In some embodiments the liquid C_2 - C_{10} product may comprise ethylene, benzene, naphthalene, and various mixtures thereof depending upon the desired products and reactions used.

The C_1 - C_3 alkane is usually heated under suitable conditions in the presence of the catalyst to produce the liquid C_2 - C_{10} product and hydrogen. Suitable conditions may vary depending upon the reactants, desired products, catalysts, and equipment employed. Typically, a temperature of from about 500, or from about 700, up to about 1000 or up to about 1200° C., and a pressure of from about 1 atmosphere up to about 3, or up to about 5, or up to about 10, or up to about 20 atmospheres may be employed to produce the liquid C_2 - C_{10} product that may comprise ethylene, benzene, naphthalene, or a mixture thereof and hydrogen. In some embodiments the C_1 - C_3 alkane is heated by burning a fuel outside the tubes in fuel burning nozzles configured to transfer heat from the burning through the tubes. As

described below, the hydrogen produced may be used as the fuel in the fuel burning nozzles outside the tubes.

Catalyst

The catalyst composition, form, size, shape, and properties may vary depending upon such parameters as the reactants, reactor type, tube size and shape, reaction conditions, and/or desired products. In some embodiments the catalyst comprises substantially cylindrical pellets. The catalyst pellets may comprise one or more holes passing through the length of the pellet, may be domed shaped on the ends, and/or may comprise a plurality of grooves on the pellet surface. Suitable catalyst pellets are described in, for example, U.S. publication US2021/0115341 to Sampath which published on Apr. 22, 2021 and is incorporated by reference herein.

In some embodiments the aspect ratio (ratio of height to diameter or cross-section length) of catalyst pellets may be at least 0.3, or at least 0.5 up to about 3, or up to about 2. In some embodiments the catalyst may comprise catalyst pellets of from about 0.5 to about 1.0, or to about 2.0 inches in diameter. In some embodiments the catalyst pellets are substantially cylindrical and wherein at least a portion of the pellets comprise one or more holes passing through the length of the pellet. The holes passing through the catalyst pellets assist in increasing the external surface area and hence generally decrease the characteristic length of the catalyst pellets to ensure adequate mass transfer rate. It may be desirable to have a characteristic length, defined as the pellet volume divided by the external surface area, to be smaller than 0.3, or smaller than 0.2, or smaller than 0.15 cm. With adequate porosity and pore structure the effective diffusivity for light hydrocarbons may be in the range of 5×10^{-3} to 2×10^{-2} cm²/s, and the catalyst effectiveness factor may be in the range of from 0.05, or from 0.1 up to 0.5, or up to 0.4.

The catalyst may comprise washcoated honeycomb monolith catalyst or metal monolith in some embodiments. The monolith may comprise ceramic, silica, quartz, glass, metal, silicon carbide, silicon nitride, boron nitride, a metal oxide or any combination thereof. Suitable metal oxides may comprise titania, iron oxide, zirconia, a mixed metal oxide, or any combination thereof.

Reactor Tubes

Like the catalyst, the tubes in the reactor may vary in shape, size, material, and/or properties depending upon such parameters as the reactants, reactor type, reaction conditions, and/or desired products. The plurality of tubes may comprise a ceramic, a metal, or a mixture thereof. Suitable metals may include, for example, alloy 800, alloy 800/HT, alloy 309, any other metallurgy suitable for high temperature services, or a mixture thereof.

In some embodiments the tubes are configured to minimize or lessen pressure drop. For example, the tubes may be configured such that a pressure drop within the plurality of tubes comprises less than about 45 psig.

If desired the plurality of tubes may comprise one or metal inserts within the plurality of tubes to facilitate the transfer of heat in a radial direction within the plurality of tubes. The metal insert may comprise a screen, a plate, or a combination thereof.

In this manner the effective thermal conductivity in a radial direction may be from about 5 to about 200 W/mK while the temperature drop may be less than 50 to 200° C. The plurality of tubes may also be configured such that the heating duty per tube is from about 2 kW/m² tube to about 70 kW/m²-tube. The size of the tubes may vary depending upon the desired heat transfer and other properties of the

system. In some embodiments, a majority of the plurality of tubes within the vessel have a diameter of from about 1, or from about 2 up to about 6, or up to about 7 inches. In some embodiments, one or more of the plurality of tubes within the vessel may have a diameter of from about 1, or from about 2 up to about 6, or up to about 7 inches.

Generally, the reactor comprises fuel burning nozzles outside and along the length of the plurality of tubes in order to burn fuel outside the plurality of tubes. The fuel is not particularly critical so long as it is capable of heating the tubes adequately. In some embodiments the fuel comprises a hydrocarbon, hydrogen, or a mixture thereof. In some embodiments the fuel may comprise hydrogen formed in the process. In some embodiments at least a portion of the heat used in heating the C₁-C₃ alkane in the tubes comprises heat from a flue gas generated during a catalyst regeneration.

FIG. 1 shows reaction temperature, pressure, and equilibrium conversion for a methane to benzene reaction according to the above embodiments.

FIG. 2 shows a representative multi-tube reactor with top fired heater. In this embodiment feed enters the tube from the top and passes over the catalyst in the tubes. The tubes comprising catalyst and feed are heated using fuel burning adjacent or near the tubes. The product of the catalytic reaction exits at the bottom of the reactor.

In the preceding specification, various embodiments have been described with references to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded as an illustrative rather than restrictive sense.

The invention claimed is:

1. A reactor for conversion of alkanes to liquid hydrocarbons and hydrogen wherein the reactor comprises:

a vessel; and

a plurality of tubes within the vessel wherein the tubes are configured to house a catalyst;

wherein the vessel is configured to burn a fuel outside the plurality of tubes and wherein the vessel is further configured to transfer heat from the burning fuel to the catalyst housed within the plurality of tubes;

wherein the reactor comprises fuel burning nozzles outside and along the length of the plurality of tubes, and wherein a majority of the plurality of tubes comprise a metal insert configured to facilitate heat transfer in a radial direction within the tubes wherein the metal insert comprises a screen, a plate, or a combination thereof.

2. The reactor of claim 1 wherein one or more of the plurality of tubes within the vessel have a diameter of from about 1 to about 7 inches.

3. The reactor of claim 1 wherein a majority of the plurality of tubes within the vessel have a diameter of from about 1 to about 7 inches.

4. The reactor of claim 1 wherein one or more of the plurality of tubes within the vessel have a diameter of from about 2 to about 6 inches.

5. The reactor of claim 1 wherein a majority of the plurality of tubes within the vessel have a diameter of from about 2 to about 6 inches.

6. The reactor of claim 1 wherein the plurality of tubes comprise a ceramic, a metal, or a mixture thereof.

7. The reactor of claim 1 wherein the plurality of tubes comprise alloy 800, alloy 800/HT, alloy 309, or a mixture thereof.

8. The reactor of claim 1 wherein the plurality of tubes configured to house catalyst are configured to house catalyst pellets of from about 0.5 to about 1.0 inches in diameter.

9. The reactor of claim 1 wherein the plurality of tubes configured to house catalyst are configured to house wash- 5 coated honeycomb monolith catalyst.

10. The reactor of claim 1 wherein the reactor comprises a feed inlet to the plurality of tubes at a top of the reactor and wherein the reactor comprises a product outlet from the plurality of tubes at a bottom of the reactor. 10

11. The reactor of claim 1 wherein the reactor comprises a fuel inlet adjacent the plurality of tubes at a top of the reactor and wherein the reactor comprises a spent fuel outlet adjacent the plurality of tubes at a bottom of the reactor.

12. The reactor of claim 1 wherein the reactor comprises 15 a feed inlet to the plurality of tubes at a top of the reactor and wherein the reactor comprises a product outlet from the plurality of tubes at a bottom of the reactor, and wherein the reactor comprises a fuel inlet adjacent the plurality of tubes at the top of the reactor and wherein the reactor comprises 20 a spent fuel outlet adjacent the plurality of tubes at the bottom of the reactor.

13. The reactor of claim 1 wherein the plurality of tubes house a catalyst for converting a C_1 - C_3 alkane to a C_2 - C_{10} product and hydrogen wherein the catalyst comprises cylin- 25 drical pellets; and

wherein at least a portion of the pellets comprise one or more holes passing through the length of a pellet, are domed shaped on ends, comprise a plurality of grooves on a pellet surface, and have an aspect ratio of at least 30 0.5 up to about 2.

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